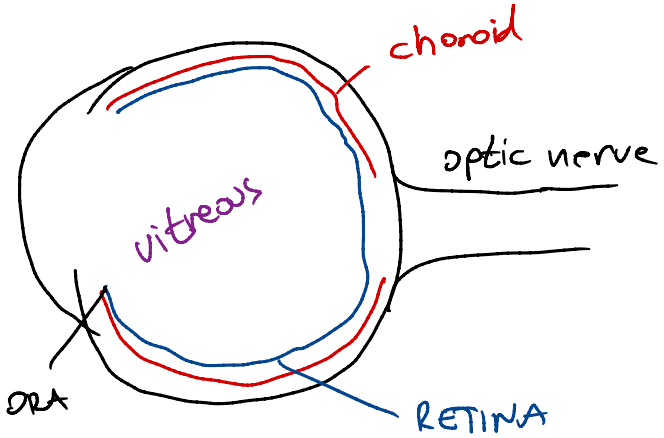
Retina one of the critical parts in sense of vision

Figures: 8.20, 8.1, 8.2, 8.4, 8.8, 8.9, 15.12, 8.32, 8.12, 8.10, 8.13, 8.27, 8.23, 8.22, 8.25, 8.26, 8.28, 8.30, 8.31

I. General retinal anatomy



A. Terms

1. Macula and fovea

central vision

2. Posterior pole

posterior portion of eyeball
retinal blood vessels, macula & ON (optic nerve)

3. Peripheral retina

anterior

4. Fundus = RETINA

B. Overview of retinal pathway different cells

main cells

1. Photoreceptors
   cones + rods
   to electrical signal
2. Bipolar cells
3. Ganglion cells
   From eye to brain

II. Retinal histology

A. Retinal pigment epithelium (RPE) outer most portion of retina

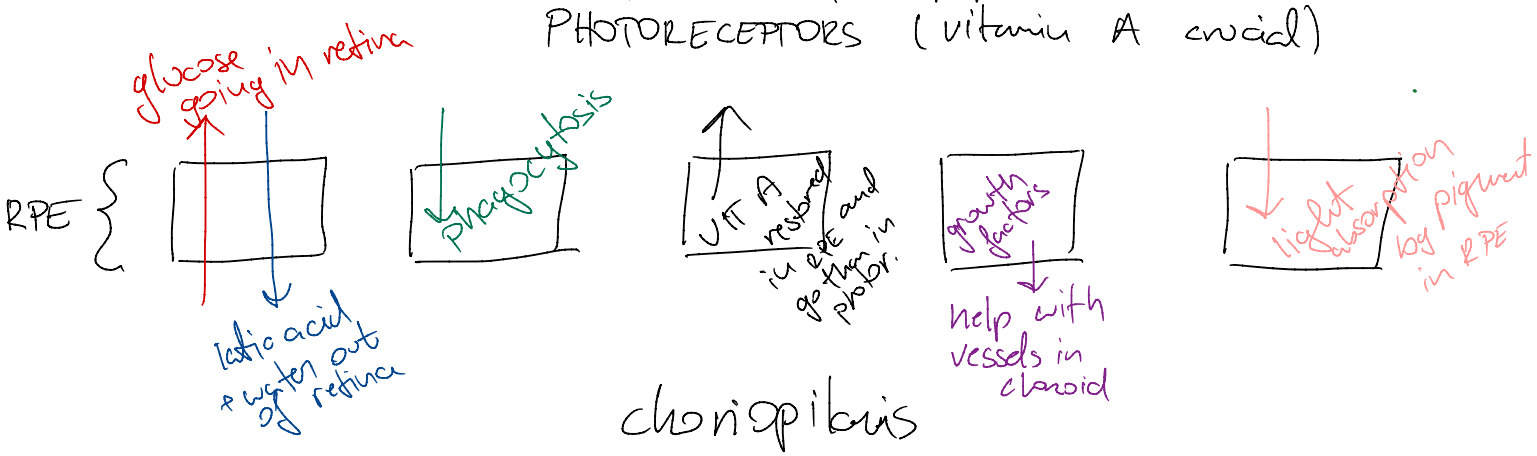
1. Single cell thick

   a. Longer, narrower, and more densely pigmented in the macular area
      that makes macula darker

2. The inner portion of the RPE, consists of microvilli that extend into the photoreceptor layer.

3. Functions of the RPE act as blood barrier

PHOTORECEPTORS (vitamin A crucial)



choriocapillaris

a. ~~Blood-retinal barrier~~ zonula occludens - very tight junction

b. Phagocytosis of photoreceptor discs

c. ~~Metabolizes and stores vitamin A~~

d. ~~Produces growth factors~~

(1) Secretes vascular endothelial growth factor (VEGF) which helps maintain choriocapillaris function

e. Absorbs light

B. Photoreceptor cells (rods and cones)

1. Contain photopigments that are activated by light

RODS → dim

CONES → BRIGHT

2. Composition of rods and cones:

a. Outer segment - contain membranous discs

rods - rhodopsin → pigment that converts into electrical signal

cones - 420nm short cone (blue)

531 nm median cones (green)

588 nm long cones (red)

b. Cilium

connecting stalk

c. Inner segment

(1) Two parts

i. Ellipsoid:

ATP → mitochondria

ii. Myoid:

inner - closer to body

contains: endoplasmic reticulum & Golgi apparatus

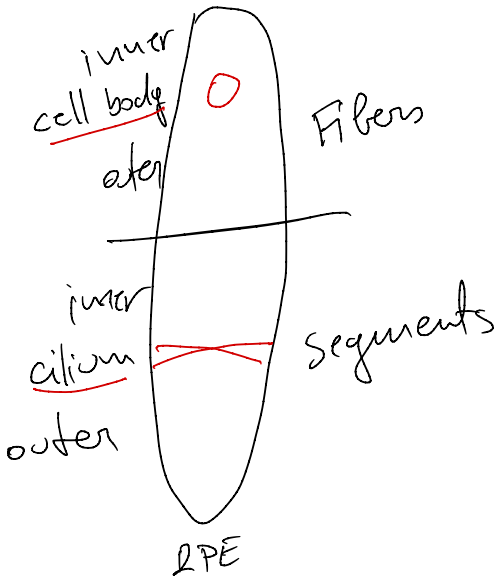
22 (Ai check)

produced to go

d. Outer fiber

e. Cell body - nucleus

f. Inner fiber - axon → synaps



3. Photoreceptors secrete the neurotransmitter glutamate

C. Bipolar cells second order neuron

1. Dendrite synapses with photoreceptors + horizontal cells
2. Axon synapses with ganglion cells + amacrine cells
   a. Secrete the neurotransmitter glutamate

D. Ganglion cells 3rd order

1. Dendrite synapses with bipolar + amacrine

2. Axon synapses with -optic radiation (= vision)
(in head not in the eye)
-super colliculus (moves my head and neck + eye to my new target → to turn when we see something)
- protectal nucleus (pupils)
- suprachiasmic nucleus (circadium rhytem)

3. Classification:

a. P-cells terminate in the parvocellular layers of the LGN Lateral "genicula" nuklcus
most important in the detail what we lookin at
(colour, shape, fine detail & contrast)

b. M-cells aka parasol ganglion cells project to the magnocellular layers of the LGN
movment and flicker

c. Intrinsically photosensitive ganglion cells (ipRGC)
By it self takes up light not through photoreceptor
they have in melanopsin
No image
- inchorge of pupil

E. Horizontal cells

1. Synapse with photoreceptors, bipolar, horizontal

2. Function

   a. Role in visual integration

   b. Modulate cone responses

F. Amacrine cells from bipolar cells to send it to ?? ? (AI check) (more outer) ganglion?

1. Synapses with bipolar, ganglion, amacrine

2. Function

   a. Modulate the information that reaches the ganglion cells

G. Muller cells ⇒ retina wouldn't "work" without them

1. Extend through much of the retina
   wraps all blood vessels

2. Function

   a. Provide structure to the retina

   b. Absorbs ions, neurotransmitters, and metabolic waste products

   c. Metabolize, synthesize, and store glycogen

III. Ten retinal layers

A. Alternate grouping of neurons and their processes

1. Retinal pigment epithelium (RPE)

2. Photoreceptor layer — outer + inner segments
muller cells

3. External limiting membrane (ELM)

4. Outer nuclear layer (ONL)
cells of rods + cones
thickest in fovea

5. Outer plexiform layer (OPL)
↳ synapse
photoreceptors
+ bipolars
+ horizontals

6. Inner nuclear layer (INL)
- horizontal cells
- bipolar
- amacrine
- muller
- deep retinal blood vessels

7. Inner plexiform layer (IPL)

bipolar + ganglion
+ amacrine
Amacrine + gang.

8. Ganglion cell layer (GCL)

thickest: surrounding the fovea

9. Nerve fiber layer (NFL) superficial blood vessels
ganglion cell axons
thickest around optic disc

a. Clinical correlate: Retinal hemorrhages

dot/blood hemorrhages are inner nuclear
flame: NFL (nerve fiber layer)

10. Internal limiting membrane (ILM)

innermost Muller cells
(portion of retina)

a. Clinical correlate: Retinal sheen

interdigitation zone (1D2)

IV. Retinal regions

A. Macula (macula lutea)

1. Darkened region in the central retina

RPE taller + more pigmented

2. Choroid is thicker than other areas of the retina

also that darken it

3. May have a yellow hue due to higher concentration xanthophyll pigments (lutein and zeaxanthin) than other areas of the retina.

a yellowish is normal, not like "staining" that is not normal

a. Function

reduce chromatic aberration

antioxidant effect

B. Fovea (fovea centralis) ONLY PHOTORECEPTORS

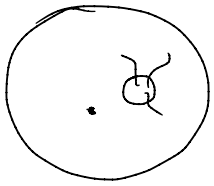
depression in center of macula

(to bipolar or ganglion, horizontal cell)

1. The bipolar and ganglion cells are displaced laterally

2. 1 mm inferior to the center of the disc

it is temporal to the disc



3. Rod free zone

4. Capillary-free zone

Not any blood vessels

C. Ora serrata

1. Peripheral termination of the retina

2. 5 mm anterior to the equator

3. Firm attachment between the retina and vitreous

D. Optic disc aka optic nerve head

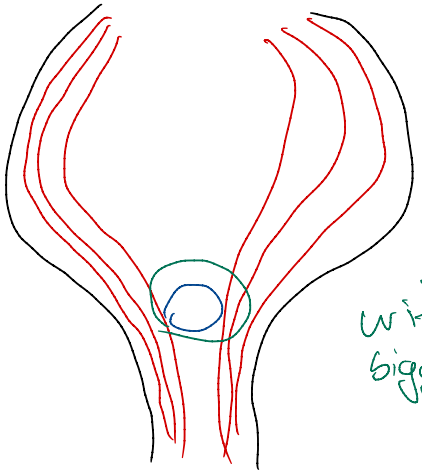
1. Ganglion cell axons accumulate and exit the eye

2. The central retinal artery enters and the central retinal vein exits the eye

3. Dimensions

   a. 1.7 mm horizontally
   b. 1.9 mm vertically

4. Physiologic cup depression



with glaucoma bigger cup

lamina cribrosa
4 little hols → deep nerve; just there, not not good or bad

V. Retinal blood supply

B. Outer retinal layers (photoreceptors) NO BLOOD VESSELS
ONLY supplied with choroid

C. Inner retinal layers
central retinal artery

D. Three capillary networks

1. Radial peripapillary capillary plexus
found in the nerv fiber layer – NFL

2. Superficial capillary plexus
GCL (AI check)

3. Deep capillary plexus
INL
border of nucleo layer

4. Outer to the outer plexiform layer is avascular.
choroid

5. Outer plexiform layer receives nutrients from both retinal and choroidal circulation.

E. Cilioretinal artery - kljukasta arterija → (Bina)

1. 15-20% of the population

2. Enters retina from the temporal edge of the disc

3. Blood supply from choroidal circulation

4. Clinical correlate: Central retinal artery occlusion